

BIHAR BOARD – 2017

Maths

Time: 3hrs 15minutes

Max Marks: 100

1. Discriminant of quadratic equation $2x^2 - 4x + 3 = 0$ is

- (a) -4 (b) 0 (c) -8 (d) None of these

Ans. -8

2. Which of the following is a rational number?

- (a) x (b) $\sqrt{7}$ (c) $\sqrt{\frac{16}{25}}$ (d) $\frac{5\sqrt{2}}{\sqrt{2}}$

Ans. $\sqrt{\frac{16}{25}}$

3. If the roots of a polynomial $x^2 - 2x + 5 = 0$ are α and β , the value of $\alpha + \beta$ will be....

- (a) -2 (b) 2 (c) -5 (d) 5

Ans. 2

4. If the first term of an A.P is a and the common difference is d, then which of the following will the n^{th} term?

- (a) $a+(n-2)d$ (b) $a+(n-1)d$ (c) $a + nd$ (d) $a -(n-1)d$

Ans. $a+(n-1)d$

5. The ratio of the sides of two similar triangles is 4:9, the ratio of their areas is

- (a) 2:3 (b) 4:9 (c) 16:81 (d) 81:16

Ans. 16:81

6. In any circle, two tangents PA and PB are drawn from any external point P. If PA = 8cm, then length of PB will be....

- (a) 4cm (b) 16cm (c) 12cm (d) 8cm

Ans. 8cm

7. $1 + \tan^2 \theta$ is equal to

- (a) $\sec^2 \theta$ (b) $\operatorname{cosec}^2 \theta$ (c) $\sin^2 \theta$ (d) $\sec \theta$

Ans. $\sec^2 \theta$

8. If a coin is tossed, the probability of getting a head will be.....

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{1}{2}$ (d) $\frac{4}{3}$

Ans. $\frac{1}{2}$

9. Mode of 2, 5, 7, 3, 3, 6 will be

- (a) 2 (b) 3 (c) 5 (d) 0

Ans. 3

10. Distance between the points (-1,3) and (-5,7) will be

- (a) $4\sqrt{2}$ unit (b) $3\sqrt{2}$ units (c) 4 units (d) 3 units

Ans. $4\sqrt{2}$ units

11. The smallest prime number is

Ans. 2

12. The x - coordinate of any point located on y-axis is

Ans. 0

13. A pair of linear equation which has no solution is known as

14. $\cos(90^\circ - \theta)$ is equal to

Ans. $\sin \theta$

15. Write 0.23 in a rational form.

Ans. $\frac{23}{100}$

16. The tangents drawn on the circumference of circle are.....

Ans. Equal.

17. If the area of a circle is $49\pi \text{ cm}^2$, then its diameter will be.....

Ans. 14cm

18. Mean of first five natural number will be.....

Ans. 3

19. If $\sin 65^\circ = a$ and $\cos 65^\circ = b$, then what will be the value of $a^2 + b^2$?

Ans. 1

20. The highest value of probability is

Ans. 1

21. Find the decimal expansion of $3\frac{3}{8}$.

Ans. $3\frac{3}{8} = 3 + \frac{3}{8} = 3 + 0.375 = 3.375$

22. If area and perimeter of a circle are same then what will be the radius of the circle?

Ans. We know,

Perimeter of circle = $2\pi r$

Area of circle = πr^2

According to the Question,

Perimeter of circle = Area of circle

$$2\pi r = \pi r^2$$

$$\text{or, } \frac{2\pi r}{\pi r} = r$$

$$\text{or, } 2 = r$$

$$\text{or, } r = 2$$

Hence the radius is 2.

23. Find the value of: $\sin 60^\circ \cdot \cos 30^\circ - \cos 60^\circ \cdot \sin 30^\circ$.

Ans.

$$\sin 60^\circ \cdot \cos 30^\circ - \cos 60^\circ \cdot \sin 30^\circ$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{1}{2} \cdot \frac{1}{2}$$

$$= \frac{3}{4} - \frac{1}{4}$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

24. Find the quadratic polynomial whose sum of zeros and product of zeros are 4 and 3 respectively.

Ans. Let the polynomial be $ax^2 + bx + c$, and its zeroes be δ and β

According to question,

Sum of zeros = 4

$$\delta + \beta = 4 = \frac{4}{1} = -\frac{b}{a}$$

And,

$$\delta \cdot \beta = 3 = \frac{3}{1} = \frac{c}{a}$$

If $a = 1$, then $b = -4$, $c = 3$

Therefore, the quadratic polynomial is $x^2 - 4x + 3$

25. A triangle ABC is an isosceles triangle whose $\angle C = 90^\circ$, prove that $AB^2 = 2AC^2$

Ans. Since $\triangle ABC$ is an isosceles triangle, $\angle C = 90^\circ$

$$\therefore AC = BC$$

Now,

By Pythagoras theorem,

$$AB^2 = BC^2 + AC^2$$

$$\Rightarrow AB^2 = AC^2 + AC^2$$

$$\therefore AB^2 = 2AC^2$$

Hence proved.

26. Find the 20th term of an A.P 9,13,17,21,.....

Ans. Given,

First term (a)=9

Common difference(d) =13-9=4

$$n = 20$$

We know,

$$t_n = a + (n-1)d$$

Now,

Putting the values, we get the

$$t_{20} = 9 + (20-1)4$$

$$= 9 + 19 \times 4$$

$$= 9 + 76$$

$$= 85$$

27. Find the co-ordinates of the mid-point that joins the point A(-5, 4) and B(7, -8).

Ans. Let P(x,y) be mid -point of AB.

Using mid- point formula, we get

$$P(x, y) = \left\{ \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right\}$$

$$= \left\{ \frac{7+(-5)}{2}, \frac{-8+4}{2} \right\}$$

$$= \left(\frac{2}{2}, \frac{-4}{2} \right)$$

$$= (1, -2)$$

28. Find the surface area of a circle whose diameter is 14cm.

Ans. Given,

Diameter (d) = 14cm

$$\therefore \text{Radius (r)} = \frac{d}{2} = \frac{14}{2} = 7\text{cm}$$

We know,

Area of circle = πr^2

$$\therefore \text{Area} = \frac{22}{7} \times 7^2 = \frac{22}{7} \times 7 \times 7 = 54\text{cm}^2$$

29. Find the co-ordinates of the point which divide the join of (-1, 7) and (4, -3) ,in the ratio 2:3.

Ans. Let P(x, y) be the required point.

Given points are (-1, 7) and (4, -3)

Ratio = 2:3 = m:n

Using the section formula, we get

$$x = \frac{2 \times 4 + 3 \times (-1)}{2 + 3} = \frac{8 - 3}{5} = \frac{5}{5} = 1$$

$$y = \frac{2 \times (-3) + 3 \times 7}{2 + 3} = \frac{-6 + 21}{5} = \frac{15}{5} = 3$$

Therefore, the point is (1, 3).

30. A side of an equilateral triangle $\triangle ABC$ is 10cm and one side of equilateral triangle $\triangle DEF$ is 5cm. Find the ratio of areas of $\triangle ABC$ and $\triangle DEF$.

Ans. Given,

Side of $\triangle ABC = 10\text{cm}$

Side of $\triangle DEF = 5\text{cm}$

We know,

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{\frac{\sqrt{3}}{4}a^2}{\frac{\sqrt{3}}{4}a^2} = \frac{(10)^2}{(5)^2} = \frac{100}{25} = 4:1$$

The required ratio is 4:1.

31. If $\sin A = \frac{5}{7}$, then find the value of $\tan A$.

Ans. Given,

$$\sin A = \frac{5}{7} = \frac{p}{h}$$

Using Pythagoras theorem,

$$\begin{aligned} b^2 &= h^2 - p^2 \\ \Rightarrow b &= \sqrt{7^2 - 5^2} \\ \Rightarrow b &= \sqrt{49 - 25} \\ \Rightarrow b &= \sqrt{24} \\ \therefore b &= 2\sqrt{3} \end{aligned}$$

Now,

$$\begin{aligned} \tan A &= \frac{p}{b} \\ \therefore \tan A &= \frac{5}{2\sqrt{3}} \end{aligned}$$

32. Use Euclid's division algorithm to find HCF of 867 and 225.

Ans. Using Euclid's division algorithm, we get

$$867 = 225 \times 3 + 192$$

$$225 = 192 \times 1 + 33$$

$$192 = 33 \times 5 + 27$$

$$33 = 6 \times 4 + 3$$

$$6 = 3 \times 2 + 0$$

$$\therefore \text{H.C.F.} = 3$$

33. Find the roots of the quadratic equation: $x^2 + 7x + 10 = 0$.

Ans. Given equation,

$$x^2 + 7x + 10 = 0$$

$$\Rightarrow x^2 + 5x + 2x + 10 = 0$$

$$\Rightarrow x(x + 5) + 2(x + 5) = 0$$

$$\Rightarrow (x + 5)(x + 2) = 0$$

$$x + 5 = 0 ; x + 2 = 0$$

$$x = -5 \quad x = -2$$

Therefore, the required root of the given equation is -5 and -2.

34. Solve the given equation by substitution method:

$$8x + 7y = 9$$

$$3x + 2y = 4$$

Ans. Given equations are

$$8x + 7y = 9 \text{-----(1)}$$

$$3x + 2y = 4 \text{-----(2)}$$

From equation (2), we get

$$3x + 2y = 4$$

$$3x = 4 - 2y$$

$$x = \frac{4 - 2y}{3}$$

Now,

Put $x = \frac{4-2y}{3}$ in equation (1)

$$8x + 7y = 9$$

$$8\left(\frac{4-2y}{3}\right) + 7y = 9$$

$$\frac{32-16y}{3} + 7y = 9$$

$$\frac{32-16y+21y}{3} = 9$$

$$32+5y = 27$$

$$5y = 27 - 32$$

$$y = \frac{-5}{5}$$

$$\therefore y = -1$$

Then, putting value of y in

$$\begin{aligned} x &= \frac{4-2y}{3} \\ &= \frac{4-2 \times (-1)}{3} \\ &= \frac{4+2}{3} \\ &= \frac{6}{3} \\ &= 2 \end{aligned}$$

Hence, the solutions of the given equation are 2 and -1.

35. Find the 31th term of an A.P whose 11th term is 38 and 16th term is 73.

Ans. Let first term of A.P be a and common difference be d.

Using the formula,

$$t_n = a + (n-1)d$$

Now ,

$$t_{11} = a + (11-1)d$$

$$\Rightarrow 38 = a + 10d \dots\dots(i)$$

Similarly,

$$t_{16} = a + 15d$$

$$\Rightarrow 73 = a + 15d \dots\dots\dots(ii)$$

Solving equation (i) & (ii), we get

$$a = -32 \text{ and } d = 7$$

Therefore,

$$t_{31} = a + 30d$$

$$= -32 + 210$$

$$= 178$$

Hence, the 31st term of the given A.P is 178.

36. Inside the triangle ABC, there is a circle which touches sides AB, BC and AC at the points D, E and F respectively. If AB = 12 cm, BC = 8cm and AC = 10cm, then find the length of CF.

Ans. Given:

$$AB = 12 \text{ cm}$$

$$BC = 8 \text{ cm}$$

$$AC = 10 \text{ cm}$$

Let us take A as external point,

Thus, AD = AF [Since, tangents from an external point are equal in length.]

$$\text{Let } AD = AF = x \text{ cm}$$

$$\text{So, } DB = 12 - x = BE \text{ [Being tangent from external point B]}$$

and

$$CE = 8 - 12 + x = -4 + x = CF \text{ [Being tangent from external point C]}$$

Now,

Since perimeter of $\triangle ABC$ does not change so

$$12 + 8 + 10 = x + x + 12 - x + 12 - x - 4 + x - 4 + x$$

$$30 = 2x + 16$$

$$2x = 14$$

$$x = 7$$

$$\therefore CF = -4 + 7 = 3 \text{ cm}$$

37. Find the co-ordinates of the point which divides the line segment joining the points A(2, -2) and B(-7, 4) into three equal parts.

Ans. Given,

$$A(2, -2) = (x_1, y_1)$$

$$B(-7, 4) = (x_2, y_2)$$

Let P and Q be the two points on the line AB so that $AP = PQ = QB$.

In that case, point C divides AB in ratio 1: 2 and point D divides AB in ratio 2 : 1

For point P; $m_1 = 1$ and $m_2 = 2$

For point Q; $m_1 = 2$ and $m_2 = 1$

Now,

For coordinates of point P ,

$$\begin{aligned} x &= \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \\ &= \frac{1 \times (-7) + 2 \times 2}{1 + 2} \\ &= \frac{-7 + 4}{3} \\ &= \frac{-3}{3} \\ &= -1 \end{aligned}$$

$$\begin{aligned} y &= \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \\ &= \frac{1 \times 4 + 2 \times (-2)}{1 + 2} \\ &= \frac{4 - 4}{3} \\ &= 0 \end{aligned}$$

For co-ordinate of point Q,

$$\begin{aligned}x &= \frac{m_1x_2 + m_2x_1}{m_1 + m_2} \\&= \frac{2 \times (-7) + 1 \times 2}{2 + 1} \\&= \frac{-14 + 2}{3} \\&= \frac{-12}{3} \\&= -4\end{aligned}$$

$$\begin{aligned}y &= \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \\&= \frac{2 \times 4 + 1 \times (-2)}{2 + 1} \\&= \frac{8 - 2}{3} \\&= \frac{6}{3} \\&= 2\end{aligned}$$

Therefore, P = (-1,0) and Q = (-4, 2)

38. A lot of 20 bulbs contain 4 defective ones. One bulb is drawn at random from the lot. What is the probability that this bulb is defective?

Ans. Given,

Total numbers of bulbs = 20

Numbers of defective bulbs = 4

$$\therefore \text{Probability of getting a defective bulb} = \frac{\text{Numbers of defective bulbs}}{\text{Total numbers of bulbs}} = \frac{4}{20} = \frac{1}{5}$$

39. Prove that : $(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}$

Ans.

$$\begin{aligned}
 & (\operatorname{cosec} \theta - \cot \theta)^2 \\
 &= \left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \right)^2 \\
 &= \left(\frac{1 - \cos \theta}{\sin \theta} \right)^2 \\
 &= \frac{(1 - \cos \theta)^2}{\sin^2 \theta} \\
 &= \frac{(1 - \cos \theta)(1 - \cos \theta)}{1 - \cos^2 \theta} \quad [\because \sin^2 \theta + \cos^2 \theta = 1] \\
 &= \frac{(1 - \cos \theta)(1 - \cos \theta)}{(1 - \cos \theta)(1 + \cos \theta)} \\
 &= \frac{1 - \cos \theta}{1 + \cos \theta} \\
 &= R.H.S
 \end{aligned}$$

40. Prove that : $\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A} = 2 \operatorname{cosec} A$

Ans:

$$\begin{aligned}
 & \frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A} \\
 &= \frac{\sin^2 A + (1 + \cos A)^2}{\sin A(1 + \cos A)} \\
 &= \frac{\sin^2 A + 1 + 2 \cos A + \cos^2 A}{\sin A(1 + \cos A)} \\
 &= \frac{1 + 1 + 2 \cos A}{\sin A(1 + \cos A)} \quad [\because \sin^2 A + \cos^2 A = 1] \\
 &= \frac{2(1 + \cos A)}{\sin A(1 + \cos A)} \\
 &= \frac{2}{\sin A} \\
 &= 2 \operatorname{cosec} A \\
 &= R.H.S
 \end{aligned}$$

41. For what value of 'x', points A(x, 2), B(-3, -4) and C(7, -5) are collinear?

Ans. Given points are A(x, 2), B(-3, -4) and C(7, -5)

Here, (x, 2) = (x_1, y_1)

$(-3, -4) = (x_2, y_2)$

$(7, -5) = (x_3, y_3)$

Since, the given points are collinear, so

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$\Rightarrow x(-4 - (-5)) - 3(-5 - 2) + 7(2 - (-4)) = 0$$

$$\Rightarrow x(-4 + 5) - 3 \times (-7) + 7(2 + 4) = 0$$

$$\Rightarrow x + 21 + 42 = 0$$

$$\Rightarrow x + 63 = 0$$

$$\therefore x = -63$$

42. In the given chart, distributions of weight of 30 students of class 'x' are shown. Find the median of the weight of students.

Weight (in kg.)	40-45	45-50	50-55	55-60	60-65	65-70	70-75
Number of students	2	3	8	6	6	3	2

Ans.

Class-interval	Frequency(f)	Cumulative frequency(c.f)
40-45	2	2
45-50	3	5
50-55	8	13
55-60	6	19
60-65	6	25
65-70	3	28
70-75	2	30
	N = 30	

$$\text{Here, } \frac{N}{2} = \frac{30}{2} = 15$$

So, the median class interval is 55-60

$$\therefore l = 55$$

$$C = 13$$

$$f = 6$$

$$i = 5$$

We know,

$$\begin{aligned} \text{Median} &= l + \left(\frac{\frac{N}{2} - C}{f} \right) \times i \\ &= 55 + \left(\frac{15 - 13}{6} \right) \times 5 \\ &= 55 + \frac{2}{6} \times 5 \\ &= 55 + 1.67 \\ &= 56.67 \end{aligned}$$

Therefore, median is 56.67 approximately.

43. A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 30° with it. The distance between the foot of the tree to the point where the top touches the ground is 8cm. Find the height of the tree.

Ans.

Let the tree be AC and is broken at B. The broken part touches at the point D on the ground.

In right $\triangle BCD$

$$\cos 30^\circ = \frac{CD}{BD}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{8}{BD}$$

$$\Rightarrow BD = \frac{16}{\sqrt{3}}$$

And

$$\tan 30^\circ = \frac{BC}{CD}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{BC}{8}$$

$$\Rightarrow BC = \frac{8}{\sqrt{3}}$$

Now,

Height of tree = BC + BD

$$= \frac{8}{\sqrt{3}} + \frac{16}{\sqrt{3}}$$

$$= \frac{8+16}{\sqrt{3}}$$

$$= \frac{24}{\sqrt{3}}$$

$$= \frac{24}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{24\sqrt{3}}{3}$$

$$= 8\sqrt{3}$$

Hence, the required height of tree is $8\sqrt{3}$ cm.

OR

From the top of the tower, a man observes two cars on the opposite side of the tower and in same straight line with its base with angles of depression 30° and 45° . If the height of the tower is 100 m, find the distance between the cars.

Ans. Let the Tower be AC = 100m

Distance between cars is BD

BD=BC+CD

Let the BC be 'x' and CD be 'y'

In triangle ACD

$$\tan 45^\circ = \frac{AC}{CD}$$

$$\Rightarrow 1 = \frac{100}{y}$$

$$\therefore y = 100m$$

In triangle ACB

$$\tan 30^\circ = \frac{AC}{BC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{100}{x}$$

$$\therefore x = 100\sqrt{3}$$

Distance between cars = BC+CD

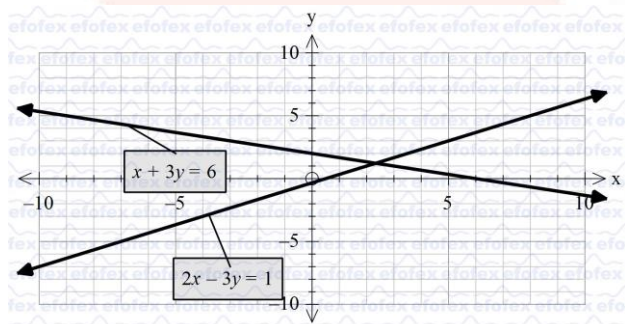
$$\begin{aligned} & 100\sqrt{3} + 100 \\ &= (100 \times 1.732) + 100 \\ &= 173.2 + 100 \\ &= 273.2m \end{aligned}$$

Hence, the required distance between cars is 273.2 m.

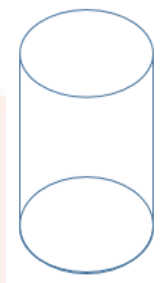
44. Solve graphically:

$$x + 3y = 6$$

$$2x - 3y = 12$$



45. An object has been made from wood by extracting a semi- shell at each end of a solid cylinder as shown in the figure. If the height of the cylinder is 12cm and radius of the base is 4.2cm, find the total surface area of the object.



Ans. Given,

Height of the cylinder (h) = 12cm

Radius of base (r) = 4.2 cm

Total surface area of cylinder = $2\pi r(h + r)$

$$\begin{aligned} &= 2 \times \frac{22}{7} \times (4.2)(4.2 + 12) \\ &= 2 \times \frac{22}{7} \times 4.2 \times 16.2 \\ &= 427.68 \text{ cm}^2 \end{aligned}$$

46. Prove that in a right angle triangle, sum of square of two sides is equal to the square of hypotenuse.

Ans. Given :-

A triangle $\triangle ABC$ in which $\angle ABC = 90^\circ$.

To prove :- $AC^2 = AB^2 + BC^2$.

Construction: - Draw $BD \perp AC$.

Proof:-

In $\triangle ADB$ and $\triangle ABC$, we have

$\angle A = \angle A$ (common).

$\angle ADB = \angle ABC$ [Both equal to 90°].

$\therefore \triangle ADB \sim \triangle ABC$ [By AA-similarity].

So,

$$\frac{AD}{AB} = \frac{AB}{AC}$$

$$\Rightarrow AB^2 = AD \times AC \dots\dots\dots(1) .$$

In $\triangle BDC$ and $\triangle ABC$, we have

$$\angle C = \angle C \text{ (common)} .$$

$$\angle BDC = \angle ABC \text{ [Both angle being } 90^\circ \text{]} .$$

$$\therefore \triangle BDC \sim \triangle ABC \text{ [By AA-similarity]} .$$

So,

$$\frac{DC}{BC} = \frac{BC}{AC}$$

$$\Rightarrow BC^2 = DC \times AC \dots\dots\dots(2) .$$

Adding equation (1) and (2) , we get

$$\Rightarrow AB^2 + BC^2 = AD \times AC + DC \times AC$$

$$\Rightarrow AB^2 + BC^2 = AC(AD + DC)$$

$$\Rightarrow AB^2 + BC^2 = AC \times AC$$

$$\therefore AB^2 + BC^2 = AC^2$$

Hence, it is proved.

47. Construct a triangle ABC in which AB = 4cm , BC = 6cm and $\angle B = 90^\circ$. Draw another similar triangle whose corresponding side is $\frac{3}{4}$ times the sides of $\triangle ABC$.

Ans. Steps of construction:

- I. Construct $\triangle ABC$ in which AB = 4cm , BC = 6cm and $\angle B = 90^\circ$.
- II. At B draw an acute angle CBX below the base BC.
- III. Along BX, mark off points B_1, B_2, B_3, B_4 on BX such that $B_1B_2 = B_2B_3 = B_3B_4$.
- IV. Join B_4 with C and draw a line through B_3 parallel to line B_4C , to intersect the line BC at C'.
- V. Draw a line through C' parallel to the line AC to intersect line BA at A'.

Thus, $\triangle A'B'C'$ is the required triangle?

OR

Draw a circle of radius 4cm. From a point 7cm away from its centre, construct a pair of tangents to the circle.

Ans. Steps of construction:

- 1) Draw a circle of radius 4 cm
- 2) Mark a point P, 7 cm away from the center B.
- 3) Draw a perpendicular bisector.
- 4) Wherever the perpendicular bisector intersect the line, take that point as center and draw a circle.
- 5) Join the points where the circle touches the circle with center B.

Hence, PQ and PR are the required tangents.